CHAPTER 1

PHYSICAL QUANTITIES

Key idea: every quantity is a number × a unit, and every term in a correct equation has the same dimensions.

1 SI BASE QUANTITIES

Quantity	Unit	Dim.
Length	metre, m	[L]
Mass	kilogram, kg	[M]
Time	second, s	[T]
Current	ampere, A	[I]
Temperature	kelvin, K	[θ]
Amount	mole, mol	[N]
Luminous intensity	candela, cd	–

Derived units are products or quotients of base units.

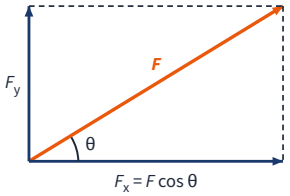
2 DIMENSIONS & HOMOGENEITY

$$[F] = \mathrm{MLT}^{-2}$$

$$[E] = \mathrm{ML}^2\mathrm{T}^{-2}$$

- A correct equation is **homogeneous**: every term has the same dimensions.
- Homogeneous does **not** mean correct: numbers such as ½ or 2π have no dimensions.
- Angles (rad), strain and refractive index are dimensionless.

3 SCALARS & VECTORS



$$F_x = F\cos\theta \qquad F_y = F\sin\theta$$

$$R = \sqrt{R_x^2 + R_y^2} \qquad \tan\theta = \frac{R_y}{R_x}$$

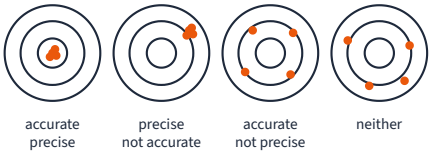
Scalar: magnitude only. **Vector:** magnitude and direction.

4 PREFIXES

Prefix	×	Prefix	×
tera T	10^{12}	centi c	10^{-2}
giga G	10^{9}	milli m	10^{-3}
mega M	10^{6}	micro μ	10^{-6}
kilo k	10^{3}	nano n	10^{-9}
deci d	10^{-1}	pico p	10^{-12}

1 cm² = 10^{-4} m², 1 cm³ = 10^{-6} m³, 1 km h^{-1} = 1/3.6 m s^{-1}

5 ACCURACY & PRECISION



Systematic — same size and sign each time (zero error); affects **accuracy**; correct by calibrating

Random — scatter both ways (reaction time); affects **precision**; reduce by averaging

6 UNCERTAINTY RULES

$Z = A \pm B$	$\Delta Z = \Delta A + \Delta B$
$Z = AB$	$\dfrac{\Delta Z}{Z} = \dfrac{\Delta A}{A} + \dfrac{\Delta B}{B}$
$Z = \dfrac{A}{B}$	$\dfrac{\Delta Z}{Z} = \dfrac{\Delta A}{A} + \dfrac{\Delta B}{B}$
$Z = A^n$	$\dfrac{\Delta Z}{Z} = \lvert n\rvert \dfrac{\Delta A}{A}$
$Z = kA$	$\Delta Z = k\,\Delta A$

- Analogue scale: ± half the smallest division
- Digital: ± one in the last digit
- Repeated: $\Delta x = \frac{1}{2}(x_{\max} - x_{\min})$

7 UNIT CHECK: DERIVED QUANTITIES

Quantity	Defining equation	SI base units	Dimensions
Force	$F = ma$	kg m s^{-2} (N)	MLT^{-2}
Energy, work	$W = Fs$	kg m^2 s^{-2} (J)	ML^{2}T^{-2}
Power	$P = W/t$	kg m^2 s^{-3} (W)	ML^{2}T^{-3}
Pressure	$p = F/A$	kg m^{-1} s^{-2} (Pa)	ML^{-1}T^{-2}
Momentum	$p = mv$	kg m s^{-1}	MLT^{-1}
Density	$\rho = m/V$	kg m^{-3}	ML^{-3}

8 COMMON SECTION A TRAPS

- Taking the newton or joule as a **base** unit; they are derived.
- Adding **percentage** uncertainties for a sum; add absolute ones.
- Forgetting to multiply by the power: $\Delta(r^2)/r^2 = 2\Delta r/r$.
- Quoting the answer to more significant figures than the data.
- Using cm in a formula that needs m.

EXAM TIPS

- ✓ Write the defining equation, then replace each quantity by its base units.
- ✓ For dimensional analysis, equate the powers of M, L and T.
- ✓ Quote the uncertainty to 1 s.f. and the value to the same decimal place.
- ✓ Resolve every vector along two perpendicular axes before adding.